

INCIDENT INVESTIGATION REPORT

INC-2026-0042 | Mechanical Seal Failure and Hydrocarbon Release, P-2104B

Incident No.	INC-2026-0042	Severity	Category 3 - Process Safety Event (Tier 2)
Date / Time of event	2026-08-29 03:40 hrs	Reported at	2026-08-29 03:52 hrs
Location	CDU-2, Grade level, Pump house bay 3	Equipment	P-2104B
Investigation lead	P. D'Souza	Report date	2026-08-31

1. SUMMARY OF EVENT

At 2026-08-29 03:40 hrs the inboard mechanical seal of P-2104B, the running CDU-2 column bottoms pump, failed and released approximately 40 litres of reduced crude oil (rco) at 341 degC to atmosphere within the pump house at Pump house bay 3. The release was detected by the field operator on routine round and by area gas detector GD-2104-1. Injuries: Nil. Fire: Nil. Area gas tested and found safe at 04:15 hrs. The unit was transferred to P-2104A in accordance with SOP-CDU2-014 Rev 4.

Production impact. Unit throughput reduced by 12% for 6 hours 20 minutes to permit changeover to P-2104A. Estimated loss 1,840 MT of crude processed.

2. SEQUENCE OF EVENTS

Date / Time	Event
2026-07-31	Monthly vibration round records 7.8 mm/s RMS on P-2104B, exceeding the 7.1 mm/s alert limit. Reading signed off with the remark 'monitor'. No corrective work order raised.
2026-08-27 23:40	Night shift handover log records oil mist and weeping at the P-2104B seal area, and an API Plan 52 reservoir level of approximately 20% against a minimum of 40%. Reservoir manually topped up with 2 litres.
2026-08-27 00:30	Handheld vibration reading of 9.4 mm/s RMS taken by the shift team. Rotating Equipment Engineer informed by telephone.
2026-08-27 01:10	Advice given to continue monitoring and keep P-2104A ready for changeover. No work order raised.
2026-08-28	Day shift monthly reading logged as 9.6 mm/s RMS. P-2104A confirmed on auto-start.
2026-08-29 03:40	Seal failure and loss of containment. Area gas detector GD-2104-1 alarms at 18% LEL.
2026-08-29 03:52	Event reported to the control room. Emergency declared at unit level. Fire water monitor coverage confirmed, no ignition.
2026-08-29 04:05	Changeover to P-2104A completed. Unit rate cut to 85% during transfer.
2026-08-29 04:15	Area gas tested clear. Emergency stood down.
2026-08-29 10:20	P-2104B positively isolated and handed over to maintenance under work order WO-26-15118.
2026-09-02	Seal replaced with kit SK-682-2104 taken from site stock. Laser alignment corrected from 0.19 mm to 0.03 mm parallel offset. Pump returned to standby.

3. IMMEDIATE CAUSE

Failure of the inboard mechanical seal (single seal, API 682 Category 1, Arrangement 1) on P-2104B, resulting in loss of containment at the seal chamber.

4. FIVE WHY ANALYSIS

#	Question	Finding
1	Why did RCO release at the pump?	The inboard mechanical seal faces lost contact and the seal chamber leaked to atmosphere.
2	Why did the seal faces lose contact?	Excessive shaft dynamic movement caused the flexible face to lift off the seat intermittently.
3	Why was shaft movement excessive?	Pump-to-driver coupling alignment had drifted to 0.19 mm parallel offset against a specification of 0.05 mm maximum.
4	Why was the misalignment not corrected?	Overall vibration crossed the 7.1 mm/s alert limit in July 2026 but no corrective work order was raised; the reading was recorded and signed off as 'monitor'.
5	Why was no work order raised on an alert-level reading?	The vibration log has no automatic escalation rule, and the monthly reliability review for July 2026 was deferred. The seal flush (API Plan 52) reservoir was also below minimum level, which was not on any routine check sheet.

5. ROOT CAUSES

- 5.1 Systemic: no mandatory escalation from an alert-level vibration reading to a corrective work order.
- 5.2 Systemic: API Plan 52 seal pot level not included in any operator or technician routine check sheet.
- 5.3 Design: single unpressurised seal (Category 1 / Arrangement 1) is marginal for 345 degC RCO service; dual pressurised arrangement is the appropriate selection.
- 5.4 Human factors: the Shift In-Charge on duty held a lapsed 'Pump Changeover - Hot Standby' competency certification (expired 2026-07-31), which extended the changeover duration.

6. EVIDENCE EXAMINED

Evidence	Observation
Failed seal faces (stationary and rotating)	Circumferential heat checking on the stationary face over approximately 60% of the contact band. Secondary sealing element hardened and lost elasticity. Consistent with intermittent dry running.
Laser alignment measurement, as found	Parallel offset 0.19 mm; angular deviation 0.11 mm per 100 mm. Specification is 0.05 mm and 0.05 mm per 100 mm respectively.
Coupling disc pack	Two of six discs showed fatigue cracking at the bolt hole. Element replaced.
API Plan 52 reservoir	Level at 22% on inspection. No level switch fitted. No routine check sheet entry exists for this item.
Vibration spectra, WO-26-11876 dated 2026-05-06	1x running speed amplitude dominant with a 2x component at approximately 40% of 1x, and axial amplitude comparable to radial. Classic misalignment signature. Report recommended alignment check; no corrective work order was raised against the recommendation.
Bearings	Thrust bearing pair showed early stage fatigue spalling on the loaded race. Replaced with stock item BRG-7314-BECBM.
Shift roster and competency records	The Shift In-Charge on duty held a 'Pump Changeover - Hot Standby' certification that expired on 2026-07-31. The roster system does not check certification validity.

7. CORRECTIVE AND PREVENTIVE ACTIONS

CAPA No.	Action	Owner	Due	Status	Remarks
CAPA-0042-1	Raise a Management of Change to upgrade P-2104A/B to API 682 Category 2, Arrangement 3 dual pressurised seals with API Plan 53B barrier fluid system.	S. Pai	2026-09-05	Complete	Raised as MOC-2026-0087 on 2026-09-01.
CAPA-0042-2	Amend SOP-CDU2-014 to require laser alignment check with 0.05 mm parallel offset tolerance recorded before returning any bottoms pump to service.	A. Fernandes	2026-09-20	In Progress	Draft Rev 5 under review.
CAPA-0042-3	Add API Plan 52 / 53B seal pot level to the operator shift round check sheet for all CDU-2 hot service pumps.	M. Shetty	2026-09-12	In Progress	
CAPA-0042-4	Configure a mandatory escalation rule: any overall vibration reading at or above 7.1 mm/s auto-generates a corrective work order within 24 hours.	P. D'Souza	2026-10-01	Not Started	
CAPA-0042-5	Re-certify all CDU-2 shift in-charges on Pump Changeover - Hot Standby and block roster assignment on lapsed certification.	A. Fernandes	2026-09-30	In Progress	2 of 6 personnel currently lapsed.

8. LESSONS AND WIDER APPLICABILITY

The failure mechanism was not sudden. An alert level vibration reading existed for a full month before the loss of containment, and a diagnostic report identifying misalignment existed for almost four months. The controlling weakness is not detection but the absence of a mandatory link between a detected exceedance and executed work. This weakness is generic to all rotating equipment on the site and is not specific to CDU-2.

A screening review is recommended for all Criticality A pumps in service above 260 degC to confirm whether the installed seal arrangement matches current engineering practice. Refer ENG-STD-SUMM-119 clause 4.1 and HAZOP action H7-04 of HAZOP-CDU2-2024.

9. INVESTIGATION TEAM AND ENDORSEMENT

Name	Role in investigation	Signature	Date
P. D'Souza	Investigation lead	Signed	2026-08-31
S. Pai	Technical member - rotating equipment	Signed	2026-08-31
N. Bhat	Technical member - process safety	Signed	2026-08-31
R. Kulkarni	Witness statement provided	Signed	2026-08-30
A. Fernandes	Endorsing authority	Signed	2026-09-01